Practice Quiz 6: Official Key

Lecture 12, Tubular Reabsorption and Secretion | Lecture 13, Concentrating and Diluting Urine

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QUIZ WINDOW	Opened **Wednesday, October 14**, after lecture. Closed **Monday, October 19**, at 11:59 pm.	OFFICE HOURS	Mon and Wed, 4:30 pm to 5:45 pm
FORMAT	10 multiple-choice items, ungraded	KEY RELEASED	**Wednesday, October 21, 2026**

Practice quizzes are ungraded self-checks and do not count toward your quiz average. Compare your answers with this key, then reread the cited slides for any item you missed. The explanations use the terms on the lecture slides.

Answer summary

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	A	D	B	A	C	B	D	D	A

Missed most often (shaded above)

1. Question 4: B, the proximal convoluted tubule. Most filtered glucose is reabsorbed there by secondary active transport with sodium.
2. Question 9: D, the principal cells of the collecting duct. ADH acts on principal cells, which insert aquaporin-2 channels.

Item-by-item key

1. Which transporter uses ATP directly and so drives most tubular reabsorption?

A The Na^+-glucose cotransporter in the apical membrane
B The Na^+-K^+-$2Cl^-$ symporter in the thick ascending limb
C The Na^+/K^+ ATPase in the basolateral membrane
D The glucose carriers for facilitated diffusion in the basolateral membrane

Answer: C. The Na^+/K^+ ATPase is the only choice that is primary active transport. It pumps Na^+ out of the tubule cell across the basolateral membrane and keeps Na^+ low inside the cell, so Na^+ enters from the filtrate down its gradient, and water, glucose, amino acids and many ions follow. Choices A and B are secondary active transport that runs on this Na^+ gradient. Choice D is passive.

Slides: L12, slides 8 and 9

2. Which statement describes the paracellular route of tubular reabsorption?

A Substances move between tubule cells, through the tight junctions.
B Substances cross the apical membrane, the cytosol and then the basolateral membrane of the tubule cell.
C Substances move from the peritubular capillaries into the tubule.
D Substances are pumped out of the tubule cell by the Na^+/K^+ ATPase.

Answer: A. The paracellular route runs between cells, through the tight junctions. In the proximal convoluted tubule these junctions are leaky, so some water and ions are reabsorbed this way. Choice B is the transcellular route, choice C is tubular secretion, and choice D is the primary active transport step of the transcellular route.

Slides: L12, slide 7; L13, slide 3

3. A person with uncontrolled diabetes mellitus has glucose in the urine. Which statement best explains this?

A Glucose is no longer filtered at the glomerulus.
B Glucose is secreted from the peritubular capillaries into the tubule.
C ADH levels are too low for glucose to be reabsorbed.
D The filtered glucose exceeds the transport maximum (Tm) of its carriers.

Answer: D. Glucose carriers are limited in number. When plasma glucose is high, the filtered glucose saturates the carriers, and the glucose above the transport maximum stays in the filtrate and appears in the urine (glycosuria). Glucose is still filtered (A), it is not secreted (B), and ADH regulates water reabsorption, not glucose (C).

Slides: L12, slides 11 and 12

4. Under normal conditions, most filtered glucose is reabsorbed in which part of the renal tubule? MISSED MOST OFTEN

A Descending limb of the nephron loop
B Proximal convoluted tubule
C Distal convoluted tubule
D Collecting duct

Answer: B. Most filtered glucose is reabsorbed in the proximal convoluted tubule by secondary active transport with sodium. Glucose crosses the apical membrane together with Na^+ and leaves the cell across the basolateral membrane by facilitated diffusion. The proximal convoluted tubule reabsorbs about 65% of the filtrate and, normally, all of the filtered glucose and amino acids. If you chose C: the distal convoluted tubule adjusts Na^+ and Ca^{2+} reabsorption under hormonal control; it does not reabsorb glucose.

Slides: L12, slides 6 and 10

5. Which hormone decreases Na^+ reabsorption, so that more Na^+ and water are lost in the urine and blood pressure falls?

A Atrial natriuretic peptide (ANP)
B Aldosterone
C Antidiuretic hormone (ADH)
D Parathyroid hormone (PTH)

Answer: A. The atria release ANP when they are stretched. ANP reduces Na^+ reabsorption, so more Na^+ and water leave in the urine, and blood volume and blood pressure fall. Aldosterone (B) increases Na^+ reabsorption, ADH (C) increases water reabsorption, and PTH (D) increases Ca^{2+} reabsorption in the distal convoluted tubule.

Slides: L12, slides 18 to 21

6. Which structure is the countercurrent exchanger, preserving the medullary osmotic gradient while it supplies blood to the renal medulla?

A Nephron loop
B Collecting duct
C Vasa recta
D Glomerular capillaries

Answer: C. The vasa recta are the countercurrent exchanger. Blood in their descending and ascending limbs exchanges water and solutes passively with the interstitial fluid, so blood flow does not wash out the gradient, which runs from about 300 mOsm in the cortex to about 1200 mOsm deep in the medulla. The nephron loop (A) is the countercurrent multiplier that builds the gradient.

Slides: L13, slides 7 to 9

7. Furosemide, a loop diuretic, increases urine output mainly because it blocks the:

A Na^+/K^+ ATPase in the basolateral membrane

B Na^+-K^+-$2Cl^-$ symporter in the thick ascending limb of the nephron loop

C Na^+-glucose cotransporter in the apical membrane

D urea transporters of the collecting duct

Answer: B. The thick ascending limb reabsorbs Na^+, K^+ and Cl^- through the Na^+-K^+-$2Cl^-$ symporter, moving NaCl into the medullary interstitium, and it is impermeable to water. This builds the medullary osmotic gradient. Furosemide blocks the symporter, so less NaCl enters the interstitium, the gradient weakens, and less water is reabsorbed from the collecting ducts. Urine output rises.

Slides: L12, slide 16; L13, slide 19

8. A substance has a renal clearance greater than the clearance of inulin. What does this tell you?

A The substance is not filtered.

B The substance is filtered and partly reabsorbed.

C The substance is filtered only, as inulin is.

D The substance is filtered and also secreted.

Answer: D. Renal clearance is C = UV / P. Inulin is filtered but neither reabsorbed nor secreted, so its clearance equals the GFR. A clearance higher than that of inulin means the tubules added more of the substance by secretion. A clearance lower than that of inulin means some of it was reabsorbed (B).

Slides: L13, slides 14 to 16

9. ADH increases water reabsorption by acting on which cells? MISSED MOST OFTEN

A Epithelial cells of the descending limb of the nephron loop

B Cells of the proximal convoluted tubule

C Intercalated cells of the collecting duct

D Principal cells of the collecting duct

Answer: D. ADH acts on the principal cells of the collecting duct, which insert aquaporin-2 channels into their apical membrane. Water then moves out of the filtrate into the hypertonic medulla, and the urine becomes concentrated. If you chose A: the descending limb of the nephron loop is always permeable to water through aquaporin-1. That is obligatory water reabsorption, and it does not depend on ADH. Intercalated cells (C) secrete H^+ or HCO_3^- for acid-base balance.

Slides: L12, slides 14, 15 and 19; L13, slide 12

10. Which statement about the urethral sphincters is correct?

A The internal urethral sphincter is smooth muscle under involuntary control, and the external urethral sphincter is skeletal muscle under voluntary control.

B Both sphincters are skeletal muscle under voluntary control.

C The internal urethral sphincter is skeletal muscle under voluntary control, and the external urethral sphincter is smooth muscle under involuntary control.

D Both sphincters are smooth muscle controlled only by the micturition reflex.

Answer: A. The internal urethral sphincter, at the bladder neck, is smooth muscle and relaxes by reflex during micturition. The external urethral sphincter is skeletal muscle, so it is under voluntary control and allows micturition to be delayed until it is convenient.

Slides: L13, slides 20 and 21

Questions about any item are welcome during office hours, Monday and Wednesday, 4:30 pm to 5:45 pm, in Science Hall S-305, or by email at rsilva@sciotocc.edu.

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